Dylan Connolly

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SUMMARY

Software engineer with experience in full-stack development, systems programming, and compiler construction. Proficient in C, C#, Python, Java, and JavaScript/TypeScript. Built an EMR platform with REST API and RBAC, an OOXML presentation editor with a parity-tested rendering engine, a Lisp compiler with optimization passes, and Linux kernel modules. Skilled with relational databases (PostgreSQL, SQLite), web frameworks (Django, ASP.NET Core, React), and DevOps tools (Git, Linux).

EDUCATION

Florida State University — Tallahassee, FL

Bachelor of Science in Computer Science, Minor in Physics, Magna Cum Laude — *July 2026*

- GPA: 3.89/4.0 · President's List: Spring 2025, Fall 2025

EXPERIENCE

Software Engineering Intern — Summer 2025

Hanku (formerly CoderOgres) — *Tallahassee, FL*

- Developed Excudo for use as an internal tool for programmatic PowerPoint generation by reverse-engineering the OOXML standard, including undocumented animation and styling specifications, cutting slide-deck production time by **32%** to accelerate curriculum work-flows.
- Engineered a text-to-speech pipeline using `chatterbox-tts` and a custom markup language, generating lecture audio locally via open-source models and removing dependence on external TTS services.
- Built PyDis, a Python bytecode disassembler with side-by-side execution visualization, variable inspection, and I/O tracing for student debugging.

PROJECTS

Excudo | ***Java 21, JavaFX, OOXML, Model Context Protocol, Python*** — Summer 2025 – Present

- Built a desktop presentation editor from scratch that reads and writes PowerPoint's native OOXML, evaluating the ECMA-376 guide formulas for all 187 preset shapes and calibrating text layout against PowerPoint's own PDF output.
- Verified rendering fidelity with an automated parity harness that scores every slide against a ground-truth PowerPoint corpus via SSIM, enforcing per-category thresholds and regression-ratcheting baselines in cross-platform CI.
- Replaced the traditional in-memory undo stack with a persistent per-slide command DAG that reconstructs edit history across restarts, and exposed the editor as an MCP server so an LLM client can inspect and edit live decks.

CliniCore | ***C#, ASP.NET Core, Entity Framework, SQLite, .NET MAUI*** — Fall 2025

- Architected EMR and practice management system with patient records, scheduling, clinical documentation, and prescription tracking across GUI and CLI interfaces.
- Implemented command pattern (CommandFactory, CommandInvoker) for unified state manipulation and undo/redo; designed medium-agnostic persistence layer supporting SQLite, in-memory, and REST API backends.
- Built complete REST API with role-based access control (RBAC) for all domain models; automated documentation with DocFX and GitHub Actions.

Lisp Compiler | ***C, Lex/Flex, Yacc/Bison, GraphViz*** — Fall 2025

- Built a complete compiler frontend for a Lisp-like language: lexical analysis, recursive descent parsing, AST generation, semantic analysis, and intermediate representation with control flow graph generation.
- Implemented optimization passes and generated DOT output for CFG visualization via GraphViz.

CounterTrak | ***Python, asyncio, aiohttp, Django, PostgreSQL*** — Spring 2025

- Built a fully asynchronous ingestion server with `asyncio` and `aiohttp` that consumes Valve's Game State Integration payloads from many concurrent Counter-Strike 2 clients, persisting up to ten snapshots per second per player into PostgreSQL.
- Designed a multi-tenant account system: users register and link Steam accounts through a Django interface, each issued a unique auth token embedded in a client-side GSI config that authenticates every payload against an in-memory token cache.
- Engineered a kill-attribution engine inferring per-weapon kills from game-state deltas, and computed player analytics over a BCNF-normalized schema using SQL window functions (`PERCENT_RANK`, `LAG`) for percentile ranking and round-economy trends.

Sysconf | ***Python, TOML, Jinja2, NGINX, Certbot, Systemd*** — 2026

- Built an idempotent Linux provisioning tool: cross-distro package management, dotfile deployment, SSH hardening, Tailscale VPN, and automated NGINX + Let's Encrypt web-app deployment.

SKILLS

Languages: C, C++, C#, Python, Java, JavaScript, TypeScript, SQL, Go, Lua, HTML, CSS, Bash

Frameworks: ASP.NET Core, .NET MAUI, Entity Framework, Django, Flask, React, Node.js, Tailwind CSS

Databases & Tools: PostgreSQL, SQLite, SQL Server, Git, GitHub, Linux, NGINX, Make, GDB, Valgrind, SSH

Concepts: OOP, Data Structures, Algorithms, REST APIs, Design Patterns, Agile, Unit Testing, CI/CD